TOPOLOGICAL INSULATORS — GRADUATE EXAM

Total Points: 50.0 | Questions: 5 | Date: March 29, 2026

Diagram-based questions on Topological Insulators

Question 1

10.0 points

Analyze the band structure diagram of a topological insulator shown in the diagram below, which features band inversion between the bulk valence and conduction bands. Explain how this inversion leads to a non-trivial topology.

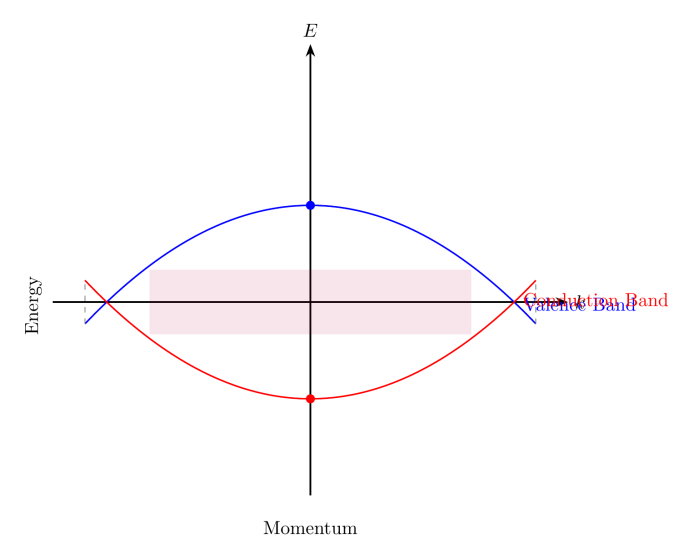

Fig. 1. A band structure diagram of a topological insulator. The diagram should show the energy bands, including the bulk valence band and conduction band, with a clear inversion between them. Highlight the band inversion that leads to a non-trivial topology. Include labels for the valence band, conduction band, and indicate the inversion point. The diagram should be suitable for illustrating the concept of band inversion in topological insulators. IMPORTANT: The diagram is missing required labels. The diagram description explicitly requires a label for the "in. Ensure ALL structural components mentioned in the description are clearly labeled. IMPORTANT: The diagram is missing required labels. The diagram description explicitly requires a label for. Ensure ALL structural components mentioned in the description are clearly labeled.

Question 2

10.0 points

Interpret the bulk-boundary correspondence as depicted in the surface state diagram of a topological insulator shown in the diagram below. Describe how the surface states are protected by topology.

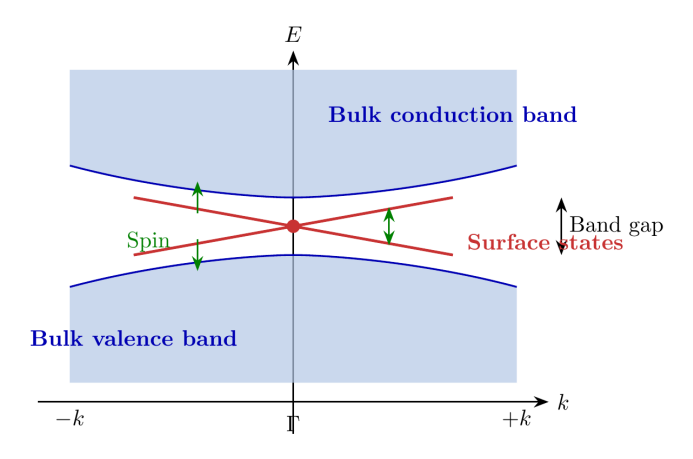

Fig. 2. A surface state diagram of a topological insulator. The diagram should show the bulk bands and the surface states crossing the band gap. The bulk bands should be represented as shaded regions, and the surface states should be depicted as lines crossing the band gap. Label the bulk bands and surface states clearly to illustrate the concept of bulk-boundary correspondence.

Question 3

10.0 points

Analyze the edge state diagram below associated with the quantum spin Hall effect. Explain how these states allow for dissipationless edge transport.

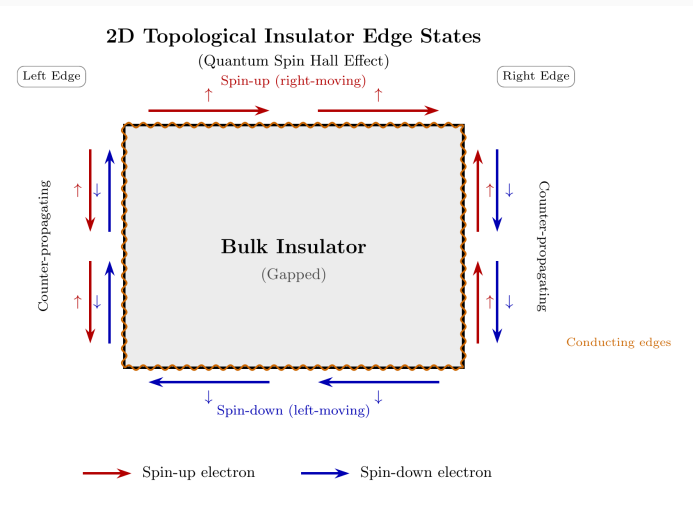

Fig. 3. An edge state diagram for the quantum spin Hall effect. The diagram should show a 2D topological insulator with edge states along the boundaries. Indicate the direction of spin-polarized edge states, with arrows showing the direction of electron flow. Highlight the distinction between the bulk insulator and conducting edges. Use labels to indicate spin-up and spin-down states moving in opposite directions along the edges.

Question 4 10.0 points

Analyze the k-space structure of a 2D system with a Z2 topological invariant as depicted in the diagram below. Explain the significance of the Z2 invariant.

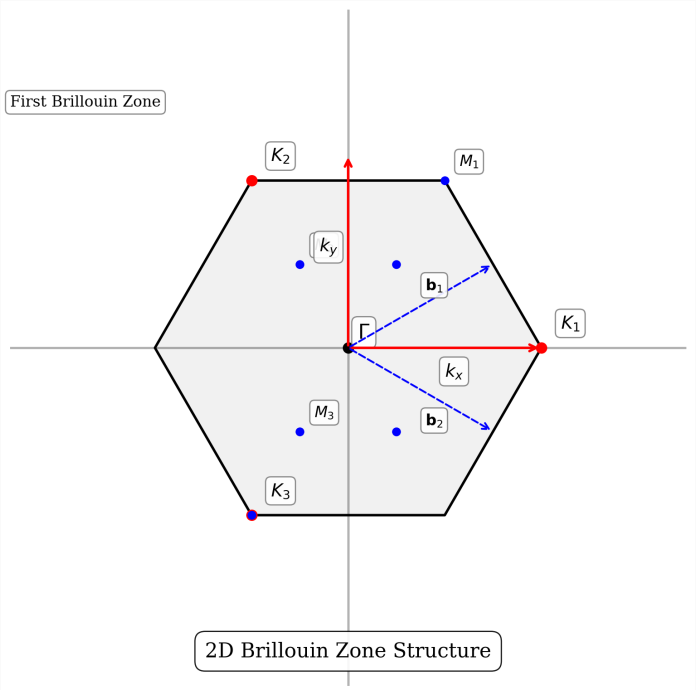

Fig. 4. A diagram of the k-space structure for a 2D system with a Z2 topological invariant. The diagram should include the Brillouin zone, with axes labeled kx and ky. Highlight any symmetry points or lines relevant to the Z2 invariant. Indicate the topological nature of the bands with appropriate annotations or shading to represent the Z2 invariant characteristics.

Question 5 10.0 points

Analyze the Dirac cone dispersion relation shown in the diagram below on the surface of a 3D topological insulator. Discuss how this structure is related to massless Dirac fermions.

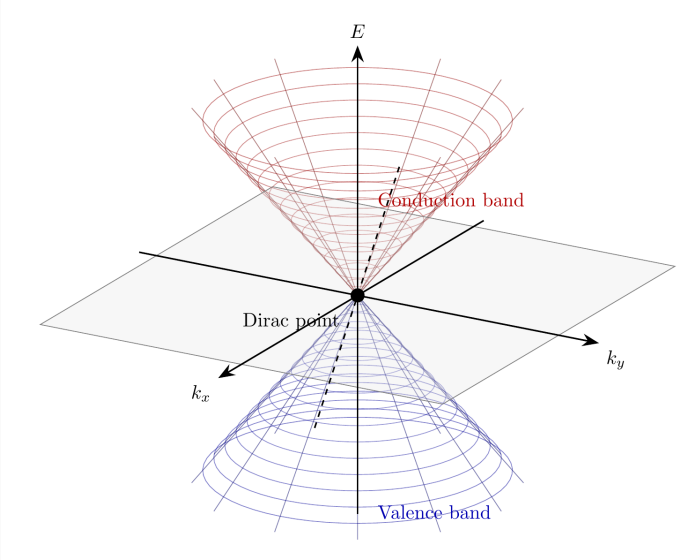

Fig. 5. A diagram showing the Dirac cone dispersion relation on the surface of a 3D topological insulator. The diagram should depict the conical energy-momentum relationship, with the energy axis (E) and momentum axes (kx, ky) clearly labeled. The cone should be centered at the origin, representing the linear dispersion of massless Dirac fermions. The surface of the cone should be smooth, indicating the continuous nature of the dispersion relation.